

A Grounded Testbed for Autonomous Scientific Discovery: Causal-Graph Memory for Self-Consistent Protein Design

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Abstract

Autonomous “AI Scientist” agents can propose and run machine-learning experiments, but two questions remain: can the full research cycle run in a *verifiable* natural-science domain, and can experimental feedback be stored so that later iterations act on it more effectively than flat text memory allows? We study both in a grounded testbed: *de novo* protein backbone design scored by self-consistency (ProteinMPNN inverse folding, ESMFold refolding, TM-align scRMSD). The only quantity the agent controls is the ProteinMPNN configuration; the scoring tools are fixed and nothing is trained, so every reported *designability* number is a real computed value. We replace EvoScientist’s flat embedding memory with a *causal failure graph* whose nodes separate measured evidence from interpretation. Running the loop with no human intervention, the agent recovers the literature-known optimum (low sampling temperature): from a deliberately far high-temperature baseline (designability 0.0) it reaches 1.0 in one improvement step, then performs one-knob ablation that leaves designability at the ceiling—growing a 42-node causal graph.

1. Introduction

Recent agentic systems automate parts of the machine-learning research loop: literature review, ideation, experiment execution, and analysis. EvoScientist (Lyu et al., 2026) organizes this as multiple agents (research, ideation, validation) that write distilled outcomes into two flat text memories retrieved by embedding cosine similarity. Two gaps motivate our work. First, most demonstrations live in ML experimentation, where the “truth” of a result is itself model-dependent; it is unclear whether the same loop closes

in a domain where each experiment returns a *verifiable* physical measurement. Second, flat text memory discards the structure of *why* an experiment failed and *what* fixed it, so a later ideation step must re-infer that structure from prose.

We address both with a single, deliberately simple system and make two contributions. **(1) Autonomy of the entire research cycle in a verifiable domain.** We test whether the full loop—literature review, hypothesis formation, experiment design, experiment execution, feedback analysis, and re-experimentation—can run with no human intervention when each experiment returns a real physical measurement rather than a model-dependent proxy. A driver re-invokes the agent on a fixed per-iteration prompt until a stopping rule fires; every iteration produces a designability score from fixed, untrained tools. **(2) Causal-graph memory that makes the next experiment better.** Where EvoScientist judges and summarizes each experiment with an LLM and writes the prose into a flat text memory retrieved by embedding similarity, we write the outcome into a typed causal graph of *attempt*, *cause*, and *fix* nodes in which *cause* nodes keep the measured numbers separate from the agent’s interpretation. Ideation then retrieves the config-nearest past attempts together with their diagnosed causes and the fixes that address them, so the structure of *why* previous runs failed—not a re-embedded paraphrase—drives the next proposal. In doing so we translate EvoScientist’s ideation and validation prompt structure into this domain and extend the paradigm from machine-learning experimentation to a verifiable scientific task. We frame the result honestly: the configuration optimum here (low temperature, best-of- N) is known from the literature; our goal is a *grounded testbed* that checks whether an autonomous loop converges to a known answer from literature plus a self-grown causal graph, and whether that graph is a better memory substrate than flat text.

Related work. The AI Scientist (Lu et al., 2024) and its successor AI Scientist-v2 (Yamada et al., 2025) automate the ML research loop end to end—ideation, coding, experiments, and paper writing—but evaluate in ML domains where success is judged by the system’s own metrics and reviewer. EvoScientist (Lyu et al., 2026) adds evolving agents with two flat text memories retrieved by cosine similarity;

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we keep its multi-agent ideation/validation structure but replace flat memory with a causal graph and move to a domain with verifiable measurements. In autonomous molecular design, ProtAgents (Ghafarollahi & Buehler, 2024) coordinates LLM agents over protein analysis and physics tools; unlike that open-ended toolkit, we fix the scoring pipeline and study whether the loop *converges* to a known configuration optimum while accumulating a reusable failure graph. Our design target itself (ProteinMPNN (Dauparas et al., 2022) scored by ESMFold (Lin et al., 2023) self-consistency (Watson et al., 2023)) is standard; the contribution is the autonomous loop and its memory, not a new design method.

2. Method

Problem and scoring device. Given a fixed single-chain protein backbone, ProteinMPNN (Dauparas et al., 2022) designs N sequences (inverse folding); each is refolded by ESMFold (Lin et al., 2023) and compared to the target backbone by TM-align (Zhang & Skolnick, 2005), yielding a self-consistency RMSD (scRMSD) and a mean pLDDT. A sequence is *designable* if $\text{scRMSD} \leq 2.0 \text{ \AA}$ and $\text{pLDDT} \geq 0.70$; *designability* is the fraction of the N sequences that pass. This self-consistency protocol is the standard designability measure in *de novo* design (Watson et al., 2023). Crucially, the scoring tools are fixed and no model is trained or fine-tuned: the *only* object the agent changes is the ProteinMPNN configuration. Every designability value is therefore a real computed measurement, which is what grounds the system against hallucinated results.

The five knobs. The agent may tune exactly five ProteinMPNN settings: the sampling temperature, the number of sequences designed per backbone, the backbone-coordinate noise, an optional set of excluded amino acids (for example cysteine), and the model checkpoint (a standard or a solubility-trained variant).¹ A one-time Research Agent web search compiles five literature insights (I1–I5) with expected directions and confidences (for example, I1: lower temperature raises designability, high confidence), used only as grounded citations for ideation.

Causal-graph memory. Memory is a directed graph with three node types—*attempt* (a scored config), *cause* (a diagnosed failure), and *fix* (a countermeasure)—and three relations *caused_by*, *fixed_by*, and *followed_by*. Retrieval, *query_relevant(config)*, ranks past attempts by knob-distance to the candidate config and returns each together with its diagnosed causes and their fixes; *to_summary_text* serializes these chains for injection into ideation. Four properties make this a better substrate

than EvoScientist’s flat, top- k embedding memory, and each follows directly from the graph structure. (i) *Relation preservation*: an attempt is retrieved with its *cause*→*fix* chain intact, so ideation sees not just that a config failed but the specific fix that addresses it; a flat store would return a paragraph from which that link must be re-inferred. (ii) *Shared-cause reuse*: because many attempts can point at the same *cause* and a *cause* can carry several *fix* nodes, a countermeasure learned once (e.g., “lower temperature”) is reusable across configs that share the failure mode, rather than duplicated per prose entry. (iii) *Hallucination resistance*: a *cause* node stores measured (the real scorer numbers) *separately* from interpretation (the agent’s hypothesis), so downstream reasoning cannot mistake a guess for a measurement—a separation flat summaries blur. (iv) *Structural retrieval*: relevance is a deterministic knob-distance over a typed key, not a lossy cosine over free text, so the nearest prior experiments are selected exactly. The *followed_by* edges additionally preserve chronology, giving a faithful trajectory of the search.

Agents and loop. Three prompt agents translate EvoScientist’s figures into this domain: an *ideation* agent (direction evolution) proposes one single-knob change, citing an insight *Ik* or a graph *fix* node; a *critique* agent scores the proposal on clarity/novelty/feasibility/relevance and enforces the hard rules (exactly one knob changed, not a duplicate, grounded, in range); a *validation* agent compares the measured designability to the running best, writes a *cause* node (measured vs. interpretation) plus 3–6 *fix* nodes on failure, and always records the outcome. A shell driver re-invokes the agent on `PROMPT.md` once per iteration; deterministic state (`STATUS.md`, `history.jsonl`, `graph_memory.json`) persists on disk, so the loop is resumable and each iteration changes at most one knob. Stopping: $\text{designability} \geq \text{target}$ or $\text{iteration} \geq \text{max}$.

3. Experiments

Setup. Backbones are single-chain PDB structures p1 (76 aa), p2 (129 aa), p3 (153 aa), and p4 (65 aa). The scorer runs on an NVIDIA A100 (CUDA); ESMFold uses fp16 with chunked attention to fit in GPU memory. We report three autonomous runs. All numbers below are read directly from `history.jsonl` produced by the loop.

Graph-guided convergence (p2). From a high-temperature baseline the loop climbs monotonically by lowering temperature one step at a time, each move citing insight I1 and the prior iteration’s *fix* node: designability $0.5 (T=1.0) \rightarrow 0.875 (T=0.7) \rightarrow 1.0 (T=0.3)$, reaching the 0.95 target in three iterations (Figure 2b).

¹Each setting maps to a verified ProteinMPNN command-line argument; the exact argument names and the cysteine and soluble-model encodings are listed in Appendix A.

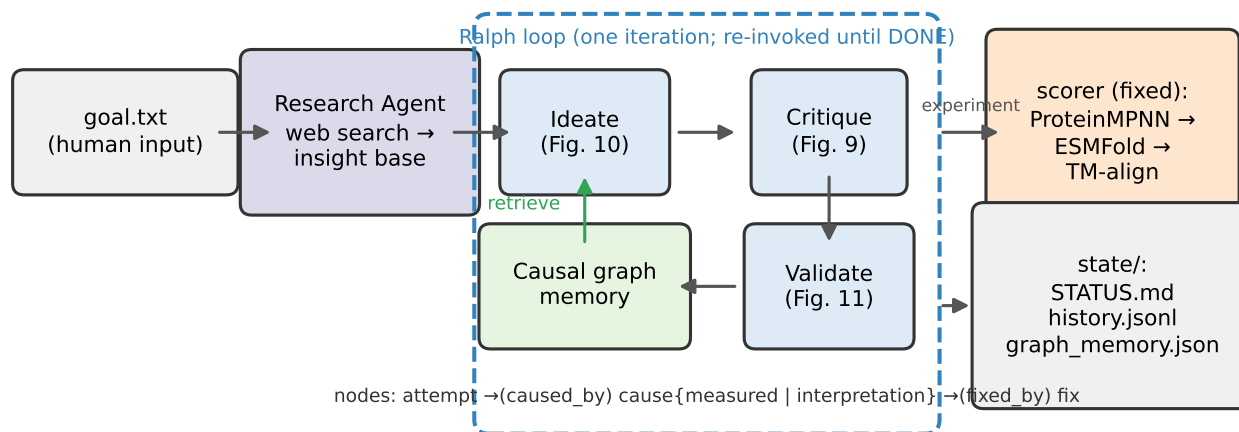


Figure 1. System architecture. A human supplies only `goal.txt`. A Research Agent performs web search once to produce a static insight base. The Ralph loop then runs one iteration per invocation—ideate, critique, experiment (the fixed scorer), validate—and reads/writes a causal graph memory and on-disk state, repeating until the stopping rule fires.

Recovery from a far baseline (p3). Started at an extreme $T=1.5$, the first iteration measures designability 0.0 (mean scRMSD 3.80 Å, mean pLDDT 0.42): sequences fail to refold. The validation agent writes a `cause` node noting that best scRMSD (2.46 Å) is already near the 2 Å threshold, so lowering temperature should suffice; the next iteration ($T=0.3$) reaches designability 1.0.

Convergence then ablation (p3, Table 1). With the target set above the ceiling to force exploration, the loop finds the optimum at iteration 2 ($T=0.1$, designability 1.0) and then spends the remaining budget on single-knob ablations around it: the soluble model, larger `num_seq`, cysteine exclusion, a low-noise checkpoint, and an even lower temperature. All hold designability at 1.0, empirically confirming that once the temperature is low, p3 is robustly designable and the remaining knobs do not change the pass rate here (they shift only the secondary scRMSD margin). Over this run the causal graph grows to 42 nodes (8 `attempt`, 7 `cause`, 27 `fix`) and 41 edges (7 `caused_by`, 27 `fixed_by`, 7 `followed_by`).

Grounding. Because the scorer is fixed and untrained, no reported number can be fabricated by the agent; the loop’s hard rules further forbid changing more than one knob per step and require `cause` nodes to keep measurements and interpretation apart. Across all runs the recovered optimum (low temperature) matches the literature prior II.

Table 1. Autonomous p3 run read from `history.jsonl`. The loop converges at iteration 2 and then performs one-knob ablation at the designability ceiling. Changed knob is relative to the running best.

it	changed knob	desig.	scRMSD	pLDDT
1	baseline ($T=1.5$)	0.000	4.462	0.39
2	$T \rightarrow 0.1$	1.000	0.856	0.89
3	model $\rightarrow$ soluble	1.000	0.936	0.89
4	num_seq $\rightarrow$ 16	1.000	0.838	0.89
5	omit_AAs $\rightarrow$ C	1.000	0.919	0.88
6	model $\rightarrow$ v_48_002	1.000	0.944	0.86
7	num_seq $\rightarrow$ 32	1.000	0.865	0.89
8	$T \rightarrow 0.05$	1.000	0.859	0.89

4. Discussion and Limitations

The system’s strength is grounding: every optimization signal is a real measurement, and the causal graph accumulates reusable, structured knowledge (here, that low temperature is decisive and the other knobs are inert at the ceiling) rather than prose to be re-embedded. Limitations follow from an honest reading. (i) Evaluation is *in silico* self-consistency, the standard designability proxy (Watson et al., 2023), not wet-lab validation. (ii) Temperature is a strong single lever, so reachable targets converge in one to two steps; interactions among the weaker knobs are only exercised in the ablation regime, and richer multi-knob landscapes would stress the memory more. (iii) Literature grounding comes from a single static search pass. (iv) We study self-consistent folding; target binding (adding an ipTM criterion) is future work. None of these affect the two claims we make: the cycle runs autonomously in a verifiable domain, and a typed

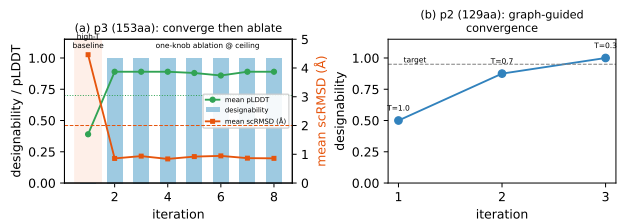


Figure 2. Real designability trajectories. (a) p3: convergence at iteration 2 followed by one-knob ablation at the ceiling; dashed lines mark the pLDDT (0.70) and scRMSD (2 Å) thresholds. (b) p2: graph-guided monotonic convergence with the temperature chosen each step annotated.

causal graph is a viable, structure-preserving memory for it.

5. Conclusion

We presented a grounded testbed in which an autonomous agent closes the full research cycle on a verifiable protein-design task and stores experimental feedback in a causal failure graph rather than flat text. With no human intervention the loop recovers the literature-known optimum and grows an interpretable memory whose cause nodes separate measurement from hypothesis. The two contributions—full-cycle autonomy in a verifiable domain, and causal-graph memory extending EvoScientist—are demonstrated end-to-end on real designability measurements.

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A. Reproducibility Details

The scorer entry point is `score_designability(backbone.pdb, config)`, returning `{backbone, config, designability, mean_scRMSD, mean_pLDDT, best_scRMSD, per_seq}`. The default config is `temperature=0.1, num_seq=8, backbone_noise=0.0, omit_AAs="", model_type="v_48_020"`.

Knob-to-argument mapping. Each of the five settings maps to a verified ProteinMPNN command-line argument: `temperature` → `--sampling-temp`; `number of sequences` → `--num_seq_per_target`; `backbone noise` → `--backbone_noise`; `excluded amino acids` → `--omit_AAs`, where the default X is always kept so excluding cysteine is passed as CX; and `model checkpoint` → `--model_name`, with a `soluble_` prefix additionally setting the `--use_soluble_model` flag to load the solubility-trained weights. Graph memory exposes `add_attempt`, `add_cause(measured, interpretation)`, `add_fix`, `query_relevant(config)`, and `to_summary_text`; the loop state manager records one `history.jsonl` line per iteration and flips STATUS to DONE when the stopping rule fires. Full agent prompts (`ideate/critique/validate`) and a `graph_memory.json` snapshot accompany the code.